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Recurrence after a First Unprovoked Cryptogenic/Idiopathic Seizure in Children: A Prospective Study from São Paulo, Brazil

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Summary: *Purpose:* To evaluate the recurrence risk after a first unprovoked seizure in a large population of children and adolescents of a developing country.

Methods: This prospective study was conducted at two tertiary hospitals, between September 1989 and August 1998. Children were enrolled if they had a first unprovoked cryptogenic/idiopathic seizure and maximal interval to the enrollment ≤90 days. EEG and computed tomography (CT) were performed in most patients. Potential predictors of recurrence were compared by using the Cox proportional hazards model in univariate and multivariate analyses. Survival analysis was performed by using the Kaplan–Meier curves.

Results: Two hundred thirteen children were included. Recurrence occurred in 34% of the patients, and mean time for recurrence was 12 months. Statistical analysis showed significance for seizure recurrence only for patients with abnormal EEGs. CT was performed in 182 patients, and abnormalities were found in 9.5%. Small calcifications were the most frequent finding, and this was not a predictor for recurrence.

Conclusions: The risk of recurrence after a first unprovoked seizure in children from a developing country is similar to that found in developed countries. An abnormal EEG is a risk factor for seizure recurrence in children with a cryptogenic/idiopathic seizure. Calcifications on CT do not increase the risk of recurrence. **Key Words:** Unprovoked seizure—Children—Recurrence rate—Risk factors—Calcifications.

The recurrence rate after a first unprovoked seizure varies widely depending mainly on the design of the study and the inclusion criteria of the subjects (1–13). Risks of recurrence vary from 26 to 71%. Retrospective studies usually show higher recurrence rates than do prospective studies.

Most studies that address this issue have been conducted in developed countries. The aim of this study was to evaluate the recurrence risk after a first unprovoked seizure in a large population of children and adolescents of a developing country and examine whether the outcome of first nonsymptomatic seizures in Brazil is similar to that seen in other countries.

METHODS

This prospective study was conducted at the Pediatric Epilepsy Clinics of two tertiary hospitals, between September 1989 and August 1998. The study protocol was submitted and approved by the ethics committees of both institutions.

Most patients were referred from emergency units of both hospitals to our Pediatric Epilepsy Clinics after the first seizure. Some patients were referred from other hospitals. Children were enrolled in the study if they had the following inclusion criteria: (a) a first unprovoked seizure defined as a seizure or a cluster of seizures all occurring in a 24-h period, without association with an acute illness, fever, central nervous system infection, trauma, or toxic or metabolic encephalopathy (8–11); (b) maximal interval to the enrollment after the initial episode ≤90 days; and (c) chronologic age between 1 month and 17 years. Exclusion criteria were (a) seizures that occurred less than a week after an insult to the central nervous system, such as trauma or infection; (b) treatment with antiepileptic drugs (AEDs) >15 days after the first episode; (c) history of previous afebrile seizures; (d) seizures that are known to have a high recurrence rate, such as absence, myoclonic seizures, infantile spasms, and symptomatic seizures.

At the time of the initial visit, patients underwent a careful history and physical and neurologic examinations, and informed consent was obtained from the parent/guardian or adolescent. A family history of seizures was defined as

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a history of an unprovoked seizure or seizures in parents or siblings. Seizure type was classified according to the International League Against Epilepsy criteria (14,15). Status epilepticus was defined as a seizure lasting >30 min or as a series of seizures, without regaining consciousness between them, lasting >30 min. Recurrence was defined as a new unprovoked seizure occurring ≥24 h after the initial seizure.

Patients had at least one EEG during the first year of the follow-up. All EEGs were performed on a 16-channel machine by using the 10-20 electrode system. All records were interictal, and almost all were obtained while the child was awake and later when the child was asleep. Photic stimulation was performed in all patients and hyperventilation when the child was cooperative. Recording time was ≥20 min. The EEG findings were classified as: normal, abnormal with nonepileptiform abnormalities, or abnormal with epileptiform abnormalities.

Computerized tomography (CT) was performed whenever possible. CT findings were classified as normal or abnormal.

At the initial visit, seizures were classified according to their etiology as cryptogenic/idiopathic or remote symptomatic (1,6,8). A seizure was classified as remote symptomatic if the child was known to have a static encephalopathy or had sustained a prior neurologic insult such as a stroke or significant head trauma. All other unprovoked seizures were considered cryptogenic/idiopathic.

Patients with unspecific findings such as arachnoid cysts, mild focal atrophy, and small calcifications were considered as having cryptogenic/idiopathic epilepsy because their causal relation could not be clearly established.

Appointments were scheduled every 6 months, unless a seizure occured. Patients were followed up for ≥1 month after enrollment. Treatment was not given to any of the children.

Analysis

The following variables were analyzed: age, gender, seizure type, number of seizures in the 24 h after the initial episode, occurrence of status epilepticus, sleep state, Todd's paresis, febrile seizure before the first unprovoked seizure, family history, CT, and EEG.

Univariate and multivariate analyses of the risk factors for recurrence were performed using the Cox proportional hazards model. Recurrence rates were determined by survival analysis, taking an event as the occurrence of a second seizure. Results are displayed as Kaplan–Meier survival curves with the cumulative probability of seizure recurrence plotted as a function of time from the first seizure. The level of statistical significance was established at $p < 0.05$.

RESULTS

The study group consisted of 213 patients, 124 boys and 89 girls. Age ranged from 1 month to 17 years (median, 9.4 years). Mean time to the enrollment after the first seizure was 19 days: 32 children were included in the study in the same day of the index seizure; 50% of the patients were included within 10 days after the initial event; 75% no later than 30 days; and 91% no later than 60 days.

The mean follow-up period was 25.7 months (1 month to 6.5 years) among patients who had no recurrence (149 subjects). One child was followed up for 1 month; two children for 2 months; all other patients were followed-up for ≥70 days; 84% of the patients were followed up for >6 months and 70% of the children for >1 year.

Table 1 shows the summary of the results obtained with univariate and multivariate analyses performed by using the Cox proportional hazards model.

Overall recurrence risk

Recurrence occurred in 73 (34%) patients, and mean time for recurrence was 12 months.

TABLE 1. *Univariate and multivariate analyses of the potential predictors of recurrence using the Cox proportional hazards model*

	Univariate (n = 213)			Multivariate (n = 138)		
Dichotomous	Rate ratio	95% Confidence interval	p Value	Rate ratio	95% Confidence interval	p Value
Abnormal EEG	2.9	1.8–4.8	0.0001	2.6	1.2–5.3	0.0001
Epileptiform			0.00002			
Nonepileptiform			0.256			
FH of seizures	1.1	0.6–2.0	0.860	1.3	0.6–2.8	>0.05
Todd's paresis	0.6	0.1–2.6	0.960	0.1	0.0–1.5	>0.05
Status epilepticus	0.8	0.3–2.3	0.943	1.4	0.4–4.8	>0.05
>1 seizure in 24 h	0.9	0.4–1.8	0.506	0.7	0.3–1.9	>0.05
Febrile seizures	0.6	0.1–1.9	0.283	0.9	0.2–3.2	>0.05
Partial seizures	1.2	0.6–2.3	0.649	1.1	0.6–2.2	>0.05
Female	1.1	0.7–1.7	0.464	1.0	0.5–1.9	>0.05
Sleep state	2.0	1.2–3.2	0.0065	1.3	0.1–2.0	>0.05
Abnormal CT	1.0	0.4–2.2	0.847	0.6	0.2–1.9	>0.05
Age (>10 yr)	0.7	0.3–1.6	0.174	0.7	0.2–2.4	>0.05

EEG, electroencephalogram; FH, family history; CT, computed tomography.

Age and recurrence

Patients were divided in three subgroups according to the chronologic age: younger than 3 years; 3–10 years; and older than 10 years. Sixteen children were younger than 3 years, 87 were between 3 and 10 years, and 110 were older than 10 years. In the first subgroup, 37% had recurrence, 39% of the children in the second subgroup had recurrence, and 30% of the adolescents had recurrence ($p > 0.05$).

Gender and recurrence

Eighty-nine were girls, and 124 were boys. Forty (32%) boys and 33 (37%) girls had recurrence ($p > 0.05$).

Type of seizure and recurrence

The type of seizure could not be established in 50 patients. Partial seizures occurred in 118 patients, and generalized seizures, in 45 subjects. Recurrence occurred in 35 (29%) patients with partial seizures and in 15 (36%) children with generalized seizures ($p > 0.05$).

Number of seizures in the first 24 h and recurrence

Patients were divided into two groups according to the number of seizures that occurred in the first 24 h after the initial seizure. Of the 213 patients, 182 had only one seizure, and 31 had 2 or more seizures. Recurrence occurred in 64 (35%) patients with one seizure and in nine (29%) children with two or more seizures ($p > 0.05$).

Status epilepticus and recurrence

Status epilepticus on the initial event occurred in 12 (6%) of the patients. Recurrence occurred in 69 (34%) patients who had no status epilepticus and in four (33%) children who had status epilepticus ($p > 0.05$).

Sleep state and recurrence

Seizures occurred in 165 awake patients and in 48 sleeping children. Recurrence occurred in 50 (30%) patients who had the initial event while awake and in 23 (48%) children who had the first seizure while asleep. In univariate analysis, the difference was significant ($p = 0.0065$), but no significant difference was found in multivariate analysis ($p > 0.05$).

Todd's paresis and recurrence

Todd's paresis occurred in six (3%) patients. Recurrence occurred in 71 (34%) patients who had no Todd's paresis and in two (33%) children with Todd's paresis ($p > 0.05$).

Febrile seizure and recurrence

Febrile seizure occurred in 14 (7%) patients. Recurrence occurred in 70 (35%) patients without febrile seizure and in three (21%) children with febrile seizures ($p > 0.05$).

Family history and recurrence

A family history of seizures was present in 45 (21.5%) patients. Recurrence occurred in 56 (34%) patients without a family history and in 16 (35%) children with a family history ($p > 0.05$).

CT and recurrence

CTs were obtained in 182 patients. Abnormal findings were detected in 17 (9.5%) children: 11 patients had lesions suggestive of neurocysticercosis (one or more nodular calcifications, and/or one or more cysts <1 cm with or without contrast enhancement); three patients had aracnoid cysts; and three patients had localized atrophy. Recurrence occurred in 64 (39%) patients with normal CTs and in seven (41%) children with abnormal CTs ($p > 0.05$).

EEG and recurrence

Abnormalities on the first EEG were found in 47 (22%) patients. The abnormalities were mainly epileptiform (38 patients). Nine patients had nonepileptiform abnormalities on their first EEGs. Recurrence occurred in 45 (27%) patients who had normal EEGs and in 28 (60%) patients with abnormal EEGs, and this was statistically significant ($p = 0.0001$). Of those with abnormal EEGs (47), 16 patients had rolandic spikes. Recurrence occurred in 10 (63%) patients with rolandic spikes on EEG and in 18 (58%) children with other types of EEG abnormalities ($p > 0.05$).

As for the multivariate Cox proportional hazards model, the Kaplan–Meier curves showed significance for seizure recurrence only for patients with abnormal EEGs (Fig. 1).

The overall Kaplan–Meier estimate of recurrence was 15% at 6 months, 25% at 12 months, 37% at 24 months, and 43% at 36 months (Fig. 2).

DISCUSSION

In this prospective study conducted at two large centers in a developing country, the risk of recurrence was similar to those reported in other prospective studies of children conducted in developed countries (8–12). The overall recurrence risk in our study was 34%, in keeping with studies that showed recurrence risks varying from 34% to 44% (1,6,8,11). The recurrence risk found in this study, which evaluated 213 children with cryptogenic/idiopathic seizures, is particularly comparable to that found by Shinnar et al. (8) when 236 children with a first idiopathic seizure had a 32% risk of recurrence.

Most studies of large series (8–11) selected patients in the emergency department. We followed up patients after the first visit in the epilepsy clinic because, in our region, pediatric neurologists usually do not see children in the emergency department. General pediatricians operate the emergency units of the two hospitals that participated in this study. They refer every child with a first unprovoked seizure to the epilepsy outpatient clinic. It may take 1 or 2 weeks for patients to be seen at the epilepsy clinic; it may take longer for patients referred from other hospitals.

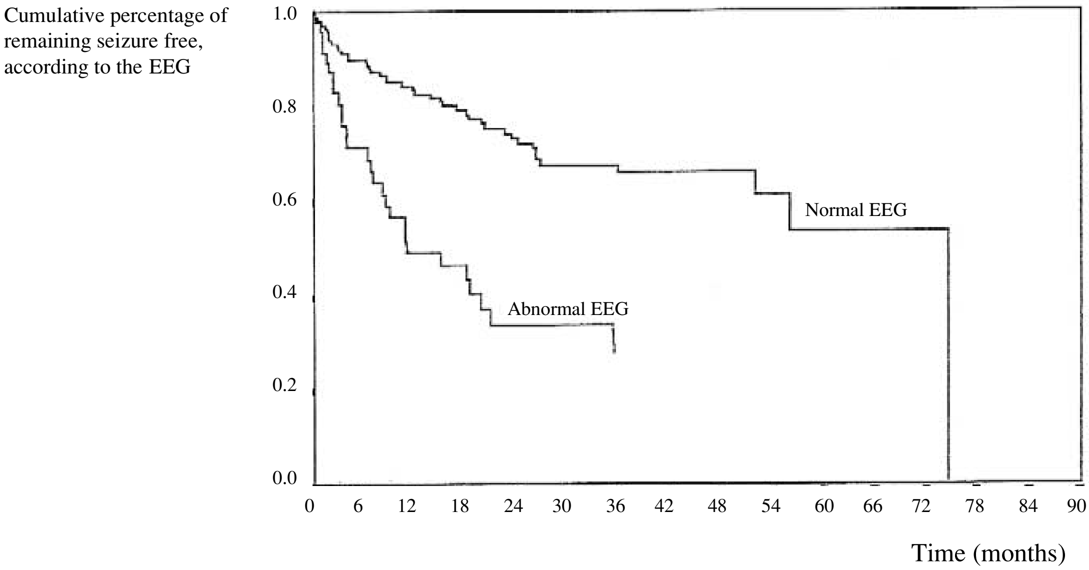

FIG. 1. Survival plot analysis (Kaplan–Meier) of the EEG finding versus seizure recurrence.

Our patients entered the study ≤90 days (mean, 19 days) after the index seizure. We may have lost some patients whose seizures recurred before the first appointment. Therefore our recurrence rate could be slightly higher if children had been selected earlier. However, both overall recurrence risk and cumulative recurrence risks in this study are in keeping with other prospective studies, particularly those that recruited patients from the emergency department (8,9,11).

The abnormal EEG was the key risk factor for recurrence in this study, and this also was found by others (8,11). Abnormalities on EEG seem to be the best predictive factor for recurrence after symptomatic etiology (12). Treatment is usually not considered after the first unprovoked cryptogenic seizure, but an abnormal EEG may influence and change this decision.

The occurrence of the first seizure during sleep was found to be a risk factor for recurrence only in univariate analysis. The relation between sleep state at the time of first seizure and recurrence was well studied by other investigators (10), who proposed two interpretations: seizures that occur in sleep may indeed differ from

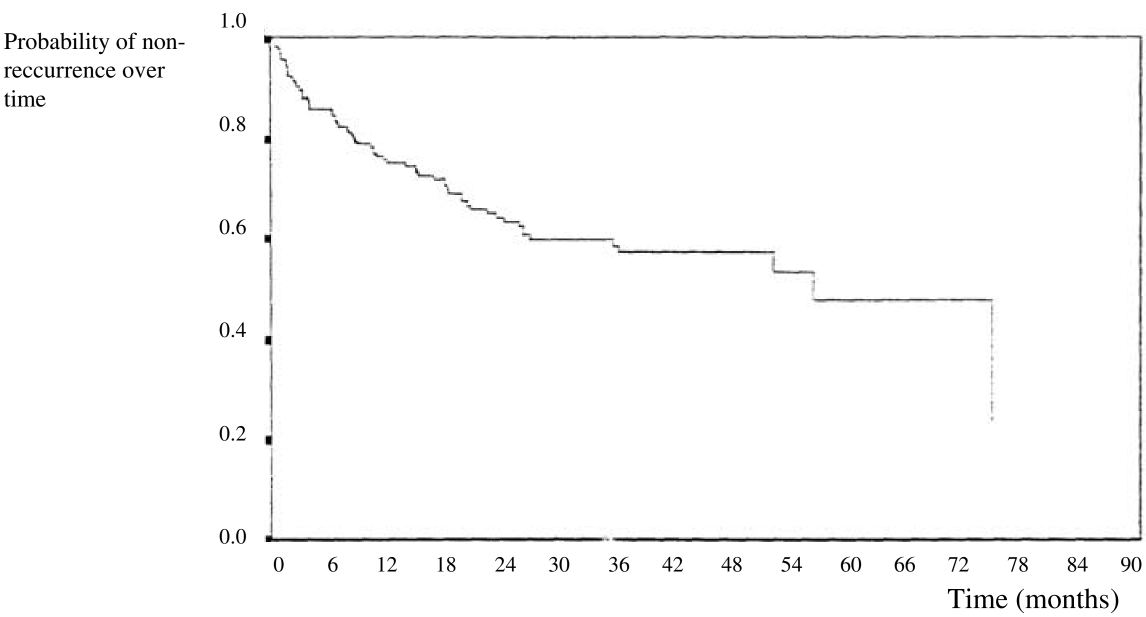

FIG. 2. Survival plot analysis (Kaplan–Meier) for seizure recurrence over a period of 3 years.

those that occur during wakefulness, or children whose first recognized seizure occurred in sleep may have had prior unrecognized seizures, which increase the risk for further seizures. In our study, we found more EEG abnormalities in the population that had the first unprovoked seizure during sleep. The association between state of sleep and abnormalities on EEG has not been completely understood. Nevertheless, the role of sleep as a modulator of cerebral activity may somehow be related to the occurrence of epileptogenic discharges on EEG.

Our data showed that age, gender, Todd's paresis, and family history were not risk factors for recurrence. Although the type of seizure did not achieve significance as a risk factor in this study, other studies suggested that a partial seizure might increase the risk of recurrence (2,3), particularly when univariate analysis is performed (8). On multivariate analysis, however, the type of seizure is not considered significant because partial seizures are more likely to occur in patients with abnormal EEGs or abnormal neurologic examinations.

The number of seizures in the first 24 h did not influence the recurrence risk in this study, as suggested by others (8,11). Nevertheless, one recent study found that if two or more unprovoked seizures occur on the same day, the child appears to have epilepsy (16).

Most of our patients underwent CT examinations, and calcifications were the most frequent abnormalities found in the small population with abnormal CTs (9.5%). Neurocysticercosis is one of the main etiologies of epilepsy in many developing countries, and calcifications are the most frequent finding on neuroimaging (17–20). The high prevalence of neurocysticercosis in our country led us to assume that the calcified lesions found on CT are cysticercosis. In our study, the risk of recurrence in the group with CT abnormalities was similar to that of the other group. The role of the calcifications in epileptogenesis is still under consideration. Some evidence suggests that at least some of them are not epileptogenic and are an incidental finding (21).

We conclude that the risk of recurrence after a first unprovoked seizure in children in a developing country was similar to that found in developed countries. An abnormal EEG is a risk factor for seizure recurrence in children with a cryptogenic/idiopathic seizure; therefore these patients should be closed followed up. Moreover, we also found that when a first seizure occurs in a child whose CT shows calcifications, it is likely that such a case will have a benign course. The small number of patients with calcifications does not allow us a final conclusion on this topic, although it suggests that small calcifications on CT should not change the decision not to treat a patient after a first unprovoked seizure.

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